a negative quantity, by definition. Naturally, the more stable a nucleus is, the more difficult it is to split it into its constituent parts, and the greater is the quantity of work that is necessary for this purpose. Therefore, stability increases with the absolute magnitude of the (negative) binding energy, and thus, strictly mathematically, the smaller the binding energy, the greater is the stability of the nucleus. For this reason, when speaking of a greater or smaller binding energy of a nucleus, we usually refer to its absolute magnitude. In this sense, the greater is the binding energy of a nucleus, the greater is its stability.

Since we are still unfamiliar with the details of nuclear structure, we are unable to calculate the binding energies from the nuclear properties. We must therefore attempt, conversely, to ascertain the magnitudes of binding energies by other methods, in order to use them as a basis for drawing conclusions about the properties of the nucleus.

The simplest example of a nucleus containing more than one particle is the deuteron, the hydrogen nucleus of mass number 2, which consists of 1 proton and 1 neutron. When such a nucleus is formed from these two component parts, the same quantity of energy must be liberated which would be necessary to break it up. Thus we start out from a state in which the proton and the neutron are still in a state of rest, far away from each other, and exert practically no force on each other. Let the quantity of energy present, under these conditions, within the system constituted by these two particles, be called 0. (The choice of zero for the potential energy of a system is arbitrary, and this choice may always be made in the most expedient manner.) As soon as the particles have combined in a deuteron, the energy of the system has decreased by the absolute magnitude of its binding energy. If we manage to measure in some way the energy content of a deuteron, we can ascertain the magnitude of its binding energy from the difference in energy before and after the joining of its constituent parts, and we can use it as a basis for reaching a conclusion concerning the stability of this nucleus.

We can proceed by the same method and add, let us say, another proton. This gives us the binding energy of the helium nucleus ${}_2He^3$. And we could continue in the same manner, step by step, to ascertain the magnitude of the binding energy of every nucleus.

The physicist customarily measures energy in *ergs* (the erg is a unit of the centimetre-gramme-second system), whereas the engineer measures it in foot-pounds or, e.g. kilowatt-hours. For thermal energies he uses the *calorie* as a unit. Thus, the different branches of physics and technical science use those units, which are most suitable for the field in question because they are designed for the orders of magnitude of energy generally encountered in that particular field, so that the energy units are neither so small nor so big as to be unwieldy or inconvenient to work with. The same principle applies to atomic